

Affine-invariant isochron curvature predicts finite-amplitude phase-reduction error in nonlinear oscillators

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Abbreviated title: Affine-invariant isochron curvature

Abstract

First-order phase reduction is intrinsically local because the leading term neglected after a finite displacement is set by curvature of the asymptotic-phase field. Direct comparison of that curvature across systems is nevertheless coordinate dependent. We formulate the local expansion in covariance-whitened state coordinates and show that every nonsingular affine reparameterization of the original state induces only an orthogonal transformation after whitening. We then test the fitted phase gradient and Hessian prospectively in 63 settings of planar nonlinear oscillators. Derivatives were estimated at 16 time-uniform phase nodes from small local perturbations and evaluated on disjoint directions at normalized amplitudes $0.025 \leq \varepsilon \leq 0.25$. Across settings, the median first-order residual scaled as $\varepsilon^{2.0017}$, whereas the residual after the Hessian correction scaled as $\varepsilon^{3.0255}$. At $\varepsilon = 0.10$, the quadratic term predicted the signed first-order residual with median slope 0.9980 and median $R^2 = 0.9834$. The Hessian reduced the setting-level median phase-reset error by 94.0921% [95% bootstrap interval 93.0536–95.2774%] at $\varepsilon = 0.10$ and by 83.7562% [81.4854–87.7703%] at $\varepsilon = 0.25$, with positive improvement in every setting. Across 250 random affine maps per setting, native and independently unit-box-normalized Hessian descriptors varied by factors as large as 5.51×10^7 and 2.17×10^{11} , respectively, whereas the largest relative discrepancy after covariance whitening was 4.92×10^{-5} . Covariance-whitened isochron curvature therefore supplies an affine-invariant local geometry that quantitatively predicts the leading finite-amplitude error of first-order phase reduction.

Keywords: asymptotic phase; isochrons; phase reduction; phase Hessian; covariance whitening; nonlinear oscillators; finite perturbations

1 Introduction

A stable limit-cycle oscillator admits an asymptotic phase function Θ whose level sets form the isochrons of the cycle [1, 2]. Restricting the dynamics to this scalar coordinate is a central reduction for synchronization, entrainment, pulse responses, and control in nonlinear systems [3, 4, 5, 6]. In the usual infinitesimal formulation, the effect of a displacement is approximated by the phase gradient on the cycle, so the quality of the reduction is controlled by how rapidly that gradient changes as the perturbation moves away from the orbit.

Higher-order phase descriptions provide the mathematical structure required beyond the infinitesimal approximation. Quadratic approximations of isochrons explicitly introduce the phase Hessian [7, 8], while phase-amplitude and isostable reductions retain transverse coordinates and extend reduced descriptions beyond the immediate neighborhood of the cycle [9, 10, 11, 12, 13, 14]. Finite pulses and higher-order interactions likewise expose corrections that are absent from a purely infinitesimal phase response [15, 16, 17]. The practical question addressed here is whether a numerically estimated local quadratic term predicts the error of first-order phase reduction on perturbations that were not used to estimate that term.

Comparisons across oscillators introduce a separate geometric problem because gradient and Hessian components depend on state coordinates. Changing voltage units, applying a shear, or linearly mixing state variables can substantially alter Euclidean norms of these derivatives without changing the underlying flow. Axis-by-axis normalization removes diagonal scale changes but does not establish invariance under a general nonsingular affine transformation. A cross-system curvature descriptor therefore requires a coordinate construction that respects affine equivalence.

The present study links those two problems by expressing the local phase expansion in covariance-whitened coordinates and then testing its finite-amplitude consequence prospectively. Orbit covariance transforms affine state changes into orthogonal changes of whitened coordinates, which preserves gradient norms, Hessian spectra, and associated orthogonally invariant descriptors. The numerical benchmark contains 48 neuronal-model settings and 15 Stuart–Landau controls, with derivative fitting and finite-amplitude evaluation performed on disjoint perturbation directions. The resulting analysis asks whether the quadratic term has the expected predictive meaning and identifies where the local approximation begins to lose quantitative accuracy.

2 Theory

2.1 Local phase expansion

Consider a smooth autonomous system

$$\dot{x} = f(x), \quad x \in \mathbb{R}^d, \quad (1)$$

with a stable hyperbolic limit cycle $\gamma(t) = \gamma(t + T)$. Phase is expressed in cycles, $\Theta \in \mathbb{R}/\mathbb{Z}$, with $\Theta[\gamma(t)] = t/T \pmod{1}$. For a displacement δx from a point on the cycle,

$$\Delta\Theta = g_x^\top \delta x + \frac{1}{2} \delta x^\top H_x \delta x + O(\|\delta x\|^3), \quad (2)$$

where $g_x = \nabla_x \Theta$ and $H_x = \nabla_x^2 \Theta$ are evaluated on the cycle. The first term is the infinitesimal phase response and the quadratic form is its leading local correction.

2.2 Covariance-whitened phase geometry

Let the time-weighted orbit mean and covariance be

$$\mu = \frac{1}{T} \int_0^T \gamma(t) dt, \quad (3)$$

$$\Sigma = \frac{1}{T} \int_0^T [\gamma(t) - \mu][\gamma(t) - \mu]^\top dt. \quad (4)$$

For full-rank Σ , define whitened coordinates by

$$y = \Sigma^{-1/2}(x - \mu), \quad (5)$$

using the symmetric positive-definite square root. Perturbation amplitudes and derivative tensors are then reported in y coordinates.

Proposition. If $x' = Ax + b$ with nonsingular A , symmetric whitening of the original and transformed cycles produces coordinates related by $y' = Qy$, where

$$Q = (A\Sigma A^\top)^{-1/2}A\Sigma^{1/2}, \quad QQ^\top = I. \quad (6)$$

Consequently, $g'_y = Qg_y$ and $H'_y = QH_yQ^\top$. Gradient norms, Hessian eigenvalues, Frobenius and spectral norms, and angles defined relative to correspondingly transformed tangent directions are invariant. The Supplemental Material gives the derivation. The result applies to nonsingular affine reparameterizations, including changes of units, rotations, shears, and linear state mixing.

3 Methods

3.1 Oscillator benchmark

The benchmark comprised 63 planar settings from five groups (Table 1). The neuronal systems were FitzHugh–Nagumo (FHN), Type-II and Type-I Morris–Lecar (ML-II and ML-I), and the Hindmarsh–Rose fast subsystem evaluated at fixed slow-variable values (HR-fast) [18, 19, 20, 21]. Stuart–Landau systems supplied analytic controls [22]. Exact equations, fixed parameters, and the complete setting list are given in the Supplemental Material.

Table 1: Benchmark composition. Each group spans a parameter lattice or list within its oscillatory regime.

Group	n	Varied parameters
FHN	18	$I \in [0.45, 1.32]$, $\tau \in \{10, 12.5, 15\}$
ML-II	11	$I \in [95, 130]$, $g_{Ca} \in [4.0, 4.8]$
ML-I	7	$I \in \{40, 45, \dots, 70\}$
HR-fast	12	$(I, \bar{z})$ in the stable-cycle region
Stuart–Landau	15	$\mu \in \{0.8, 1.1, 1.4\}$, $\beta \in [0, 2]$

3.2 Cycle extraction and time phase

Numerical limit cycles were obtained with adaptive DOP853 integration after transient rejection. Consecutive maxima of the first state variable defined the section used to estimate the period, and each accepted orbit was stored at 4096 points uniformly spaced in elapsed time. Acceptance required finite states, nontrivial amplitude, stable section intervals, and normalized closure error below 10^{-5} . The nontrivial planar Floquet log multiplier was also estimated from $\int_0^T \nabla \cdot f[\gamma(t)] dt$. Stuart–Landau cycles and phases were evaluated analytically.

3.3 Local derivative fitting and held-out perturbations

At each of 16 equally spaced phase nodes, the local fitting cloud contained 16 angular directions at radii 0.01 and 0.02 in whitened coordinates, for 32 local samples per node. A least-squares fit with no intercept used the design

$$[\delta y_1, \delta y_2, \frac{1}{2}\delta y_1^2, \delta y_1\delta y_2, \frac{1}{2}\delta y_2^2] \quad (7)$$

to estimate g_y and the symmetric matrix H_y . Evaluation used a different angular set: 12 directions with a fixed offset from the fitting directions were tested at $\varepsilon \in \{0.025, 0.05, 0.10, 0.15, 0.20, 0.25\}$. The design therefore produced 1152 held-out evaluation trials per setting (Fig. 1).

Benchmark geometry and prospective derivative-testing design

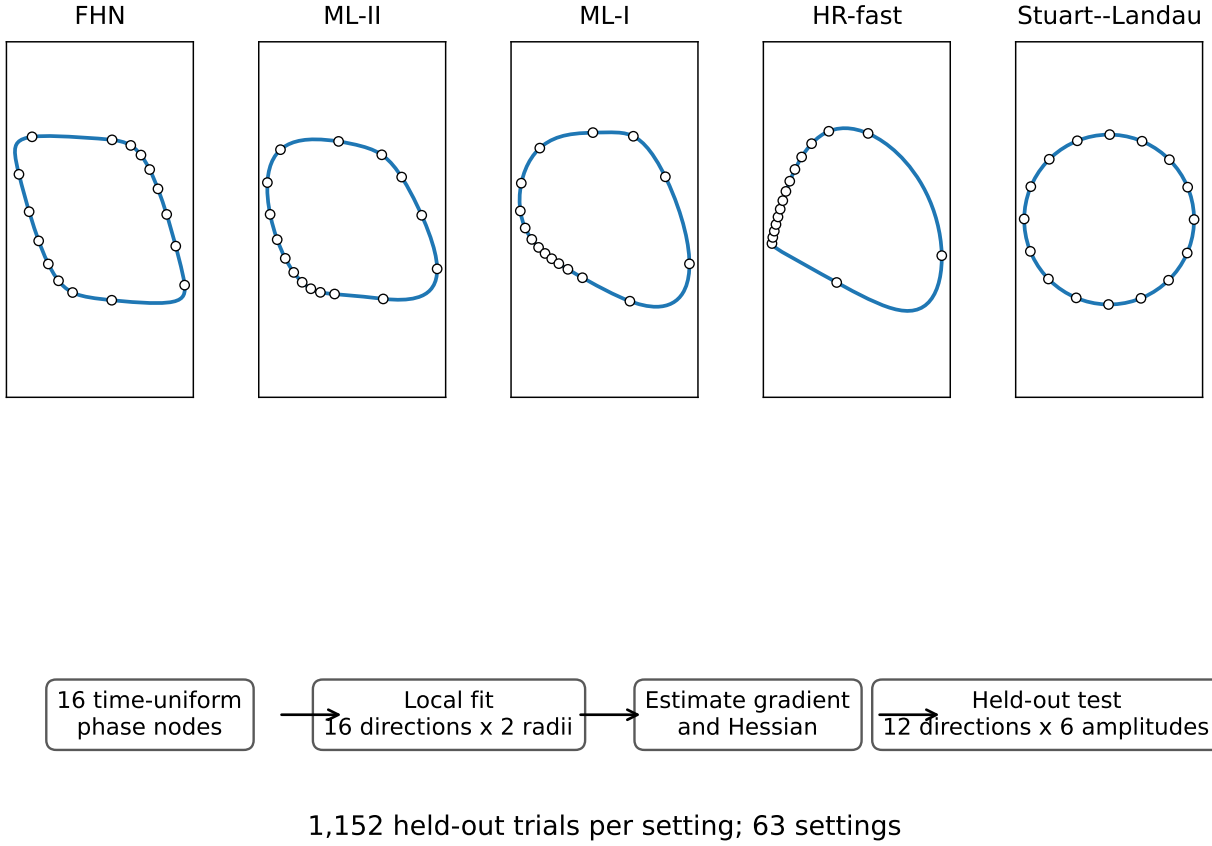


Figure 1: Benchmark geometry and prospective derivative-testing design. Representative limit cycles are shown in covariance-whitened coordinates with the 16 time-uniform phase nodes indicated by open circles. The local derivative fit and finite-amplitude test use disjoint angular direction sets.

3.4 Reference phase resets and prediction error

For FHN, ML, and HR-fast settings, perturbed states and their phase-node reference states were integrated together for two periods using vectorized fourth-order Runge–Kutta integration. The maximum time step was 0.05 for FHN, 0.12 for ML, and 0.0125 for HR-fast. Final states were

projected onto a periodic cubic-spline representation of the cycle in whitened coordinates, with the corresponding reference states carried through the same integration so that common numerical phase drift cancels. Stuart–Landau phase resets were evaluated from the exact phase

$$\Theta(r, \theta) = \frac{\theta + \beta \ln(r/\sqrt{\mu})}{2\pi} \pmod{1}. \quad (8)$$

A numerical trial was accepted when its final whitened distance from the cycle was below 5×10^{-3} and all phase quantities were finite.

The first- and second-order predictions were

$$\Delta\Theta_1 = g_y^\top \delta y, \quad (9)$$

$$\Delta\Theta_2 = g_y^\top \delta y + \frac{1}{2} \delta y^\top H_y \delta y. \quad (10)$$

Errors were absolute circular differences in cycles. Error statistics were first computed within each setting and amplitude and then summarized across the 63 settings. Ninety-five-percent intervals were nonparametric percentile bootstrap intervals from 10,000 setting-level resamples. For $\varepsilon \leq 0.10$, log–log slopes of median error versus amplitude were estimated separately for each setting. At $\varepsilon = 0.10$, the signed first-order residual was regressed through the origin on the fitted quadratic term, which directly tests whether the Hessian predicts both sign and magnitude of the omitted contribution.

3.5 Affine-coordinate stress test and numerical validation

For each setting, 250 nonsingular affine maps were formed from two independent rotations and singular values sampled log-uniformly from 10^{-2} to 10^2 , followed by translation. Numerically fitted native tensors and cycles were transformed algebraically, after which native, independent unit-box, and covariance-whitened descriptors were recomputed. The resulting 15,750 transformations isolate the coordinate effect from dynamical reintegration.

Numerical validation combined exact Stuart–Landau derivative comparisons, a halved-integration-step rerun of eight representative numerical settings, separate fits at radius 0.01 or 0.02, and even-versus-odd phase-node subsets. These tests quantify derivative accuracy, integration sensitivity, local-fit-radius sensitivity, and phase-sampling sensitivity without changing the primary held-out endpoint.

4 Results

4.1 Cycle and derivative accuracy

All 63 settings satisfied the cycle criteria, with periods ranging from 1.653 to 943.739 model-time units and a maximum normalized closure error of 1.13×10^{-7} . The analytic Stuart–Landau controls reproduced both derivative orders across 240 phase nodes: the maximum relative error was 2.273×10^{-14} for the gradient and 8.607×10^{-12} for the Hessian. Halving the integration step in representative FHN, ML-II, ML-I, and HR-fast settings changed the 95th-percentile phase reset by at most 8.658×10^{-8} cycles and the error-reduction fraction by at most 6.623×10^{-6} .

4.2 The Hessian predicts the leading truncation error

The first-order residual followed the expected quadratic amplitude law, with a median setting-level log-log exponent of 2.0017 [95% interval 1.9983–2.0044]. After inclusion of the Hessian, the median exponent increased to 3.0255 [3.0016–3.1086], consistent with a remaining cubic-order term (Fig. 2). At $\varepsilon = 0.10$, the quadratic term predicted the signed first-order residual with median through-origin slope 0.9980 [0.9974–1.0000], median $R^2 = 0.9834$ [0.9687–0.9920], and median correlation 0.992. The fitted quadratic form therefore recovered the independently observed residual rather than merely reducing an unsigned error measure.

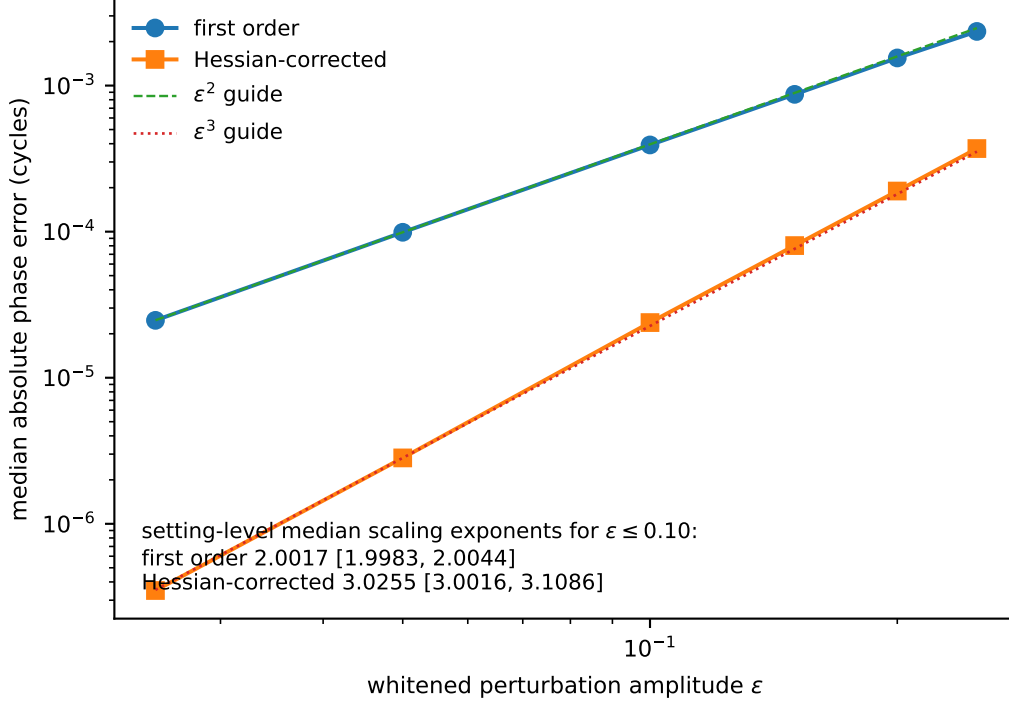


Figure 2: Finite-amplitude truncation error. Points show medians across setting-level median absolute phase errors. The reference guides are proportional to ε^2 and ε^3 ; the annotated intervals summarize the distribution of setting-level scaling exponents over $\varepsilon \leq 0.10$.

4.3 Prospective error reduction is retained across all settings

The Hessian correction reduced median phase-reset error at every tested amplitude (Table 2). At $\varepsilon = 0.10$, the reduction was 94.0921% [93.0536–95.2774%], and at $\varepsilon = 0.25$ it remained 83.7562% [81.4854–87.7703%]. All 63 settings showed positive setting-level median improvement at both amplitudes (Fig. 3). Coverage was complete through $\varepsilon = 0.15$; the minimum was 188/192 at $\varepsilon = 0.20$ and 186/192 at $\varepsilon = 0.25$. Eleven of the 72,576 evaluation trials failed the final-distance criterion, concentrated in one low-drive ML-II setting and one low-effective-drive HR-fast setting.

The strongest departure from near-perfect residual prediction occurred in the ML-I setting at $I = 40$, immediately above the SNIC onset in the chosen parameterization. Its period was 943.739, and the Hessian reduced median error by 75.8% at $\varepsilon = 0.10$ but by 30.6% at $\varepsilon = 0.25$. The signed-residual regression gave slope 0.785 and $R^2 = 0.581$. The correction remained beneficial,

Table 2: Prospective phase-reset errors. Errors are medians across the 63 setting-level medians. Brackets give 95% setting-bootstrap intervals for the reduction; coverage is the minimum valid fraction among settings.

ε	First order	Hessian-corrected	Reduction (%)	Min. coverage
0.025	2.471×10^{-5}	3.495×10^{-7}	98.7 [98.4, 98.8]	1.0000
0.050	9.902×10^{-5}	2.826×10^{-6}	97.3 [96.4, 97.7]	1.0000
0.100	3.923×10^{-4}	2.386×10^{-5}	94.1 [93.1, 95.3]	1.0000
0.150	8.708×10^{-4}	8.030×10^{-5}	91.3 [88.8, 92.9]	1.0000
0.200	1.546×10^{-3}	1.897×10^{-4}	88.0 [84.7, 90.3]	0.9792
0.250	2.348×10^{-3}	3.706×10^{-4}	83.8 [81.5, 87.8]	0.9688

while the deterioration of the local residual relation identifies a quantitative transition toward a regime where a phase-amplitude description is more appropriate.

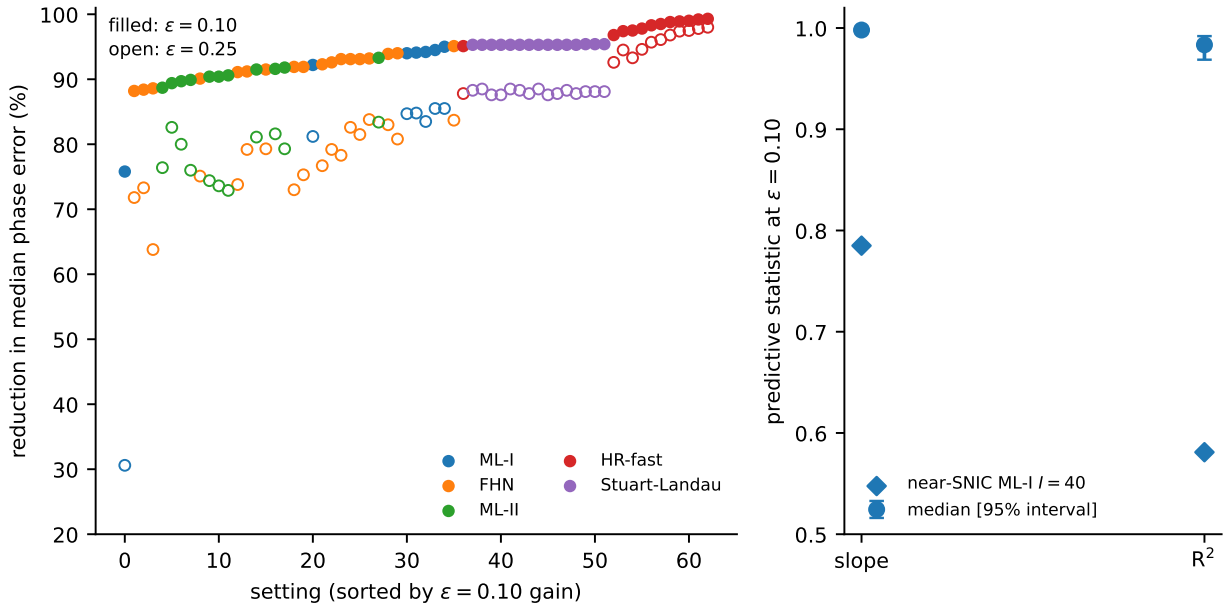


Figure 3: Setting-level predictive results. Left: reduction in median phase-reset error at $\varepsilon = 0.10$ and 0.25 for all 63 settings, ordered by the $\varepsilon = 0.10$ gain. Right: median signed-residual regression statistics at $\varepsilon = 0.10$ with 95% setting-bootstrap intervals, together with the near-SNIC ML-I boundary setting.

4.4 Covariance whitening removes affine coordinate artifacts

Affine reparameterization changed native gradient norms by as much as a factor of 7.072×10^3 and native Hessian norms by 5.508×10^7 . Independent unit-box normalization did not remove the coordinate dependence: the corresponding maximum spreads reached 4.266×10^5 for the gradient and 2.171×10^{11} for the Hessian. Covariance whitening reduced each affine state transformation to orthogonal equivalence, and the largest relative discrepancy among the tested whitened gradient and Hessian descriptors was 4.919×10^{-5} even though affine-map condition numbers approached

4.838×10^3 (Fig. 4).

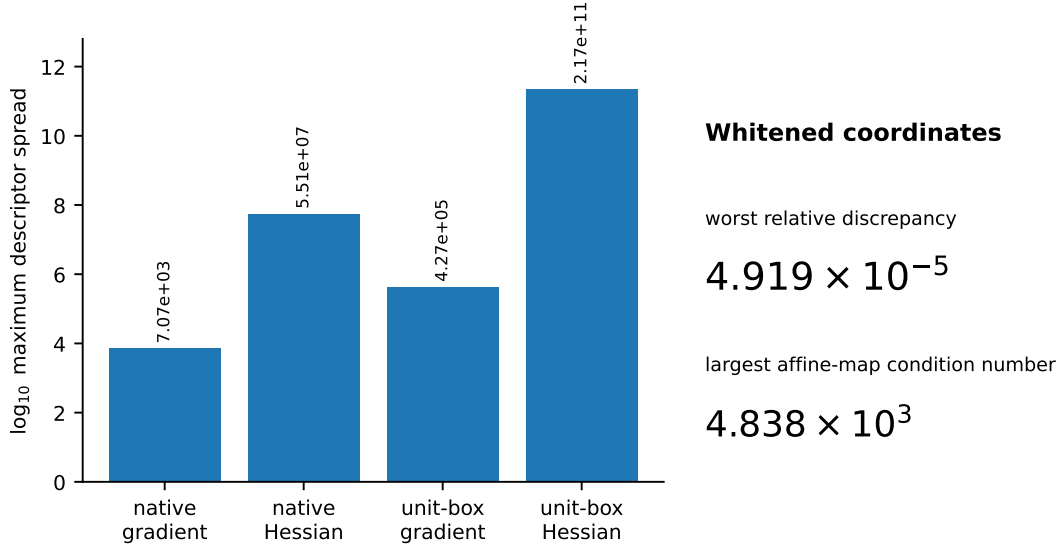


Figure 4: Affine-coordinate stress test. Native and independently unit-box-normalized derivative descriptors show large spreads across 250 affine maps per setting. Covariance-whitened descriptors remain equivalent up to the numerical discrepancy shown at right.

5 Discussion

The central result is a prospective relationship between local phase geometry and finite-amplitude reduction error. A Hessian estimated from small perturbations predicted the signed error made by first-order phase reduction on different directions and larger amplitudes. The observed ε^2 and ε^3 error laws, the near-unit residual slope, the high setting-level R^2 , and the positive error reduction in every setting form a coherent quantitative signature of the Hessian as the leading finite-amplitude correction.

This result extends established second-order phase theory by adding a coordinate-robust comparison layer and a held-out numerical test across heterogeneous oscillator families. Quadratic isochron approximations were developed previously [7, 8], and phase-amplitude or isostable coordinates provide richer descriptions of transient dynamics [11, 12, 13, 14]. The contribution here is to connect those local derivatives to an explicit finite-perturbation prediction while removing affine coordinate arbitrariness. That connection also supplies an operational criterion for model reduction: the phase-only quadratic description is quantitatively useful while its predicted residual remains aligned with the observed first-order error, whereas a loss of that relation signals the need for an amplitude coordinate.

The affine construction is especially relevant to dynamical models whose state variables carry different physical units or whose observed coordinates are linear mixtures of underlying variables. Diagonal nondimensionalization can remove known axis scales, but shear and general linear mixing remain free to alter derivative norms. Covariance whitening removes that ambiguity for nonsingular affine transformations by reducing them to orthogonal equivalence. Physical perturbation channels retain their interpretation because predictions can be mapped between whitened and native

coordinates through the same linear transformation.

The near-SNIC setting shows how the criterion behaves at a demanding dynamical boundary. Its very long period and strongly nonuniform phase progression coincide with a marked reduction in large-amplitude accuracy, while the second-order term still improves the phase-reset estimate. Slow-manifold phase-amplitude methods provide a natural next representation for this regime [14]. More generally, the setting-level results separate a gradual loss of local quantitative accuracy from a categorical failure of the second-order correction.

The present benchmark addresses smooth, deterministic, autonomous planar oscillators. The HR analysis uses a fixed slow variable and therefore represents the fast subsystem rather than a complete bursting trajectory. Hybrid reset models require saltation-aware phase theory [23], while stochastic forcing, higher-dimensional conductance-based systems, and non-affine coordinate transformations call for distinct extensions of the present construction. Covariance whitening also presumes full-rank orbit covariance. These domains define direct tests of how far the affine-invariant local geometry can be generalized beyond the planar deterministic setting studied here.

6 Conclusion

Covariance whitening converts nonsingular affine state reparameterizations into orthogonal changes of local phase coordinates, making norm- and spectrum-based curvature measures directly comparable across systems. Within that geometry, the fitted phase Hessian prospectively predicts the leading finite-amplitude error of first-order phase reduction across 63 oscillator settings, with the expected quadratic-to-cubic change in residual scaling and positive error reduction in every setting. The residual relation also exposes the onset of a validity boundary near a SNIC regime, providing a quantitative basis for deciding when a phase-only approximation should give way to a phase-amplitude description.

Data and code availability

The setting-level numerical tables underlying the figures and reported summary statistics, together with the script used to regenerate the figures, accompany the Supplementary Material. The complete computational source and trial-level benchmark are available from the corresponding author.

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